PHISHING EMAIL DETECTION AND AWARENESS REPORT

CYBER SECURITY TASK 2

FUTURE INTERNS (2026)

Tinashe Nyakumbi

1. Introduction

Phishing is a cyber attack where attackers impersonate legitimate organizations to trick users into revealing sensitive information such as passwords, banking details, or personal data.

These attacks rely on social engineering techniques such as urgency, fear, and deception rather than exploiting technical vulnerabilities.

This report presents an analysis of phishing email samples, identifies key indicators, classifies risks, and provides user awareness guidelines.

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2. Objective

The objectives of this report are:

- Analyze phishing email samples
- Identify phishing indicators
- Classify email risks
- Explain phishing attacks in simple terms
- Provide prevention and awareness guidelines

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3. Tools Used

- Google Message Header Analyzer
- MXToolbox Email Header Analyzer
- Web browser tools for safe inspection
- Public phishing datasets

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4. Email Sample Analysis

Sample 1: Account Verification Phishing Email

Full Email Header

From: security@fakebank.com
To: user@example.com

Subject: Urgent, Your Account Will Be Locked
Date: Tue, 15 Oct 2024 19:45:40 +0000

Full Email Body

Dear User,

We noticed suspicious activity on your account.
To avoid account suspension, please verify your details immediately.

http://secure-account-verify.com

Failure to verify within 24 hours will result in permanent account lock.

Regards,
Security Team

Phishing Indicators:

- Generic greeting ("Dear User")
- Urgent and threatening language
- Suspicious and non-official URL
- Lack of personalization
- No verifiable sender identity

Risk Classification- PHISHING (High Risk)

Explanation

This email uses fear-based tactics to pressure the user into clicking a malicious link.
The link likely redirects to a fake login page where user credentials are stolen.

Sample 2: Obfuscated Financial Scam Email

Full Email Header

From: natloulcinjd@hotmail.com
To: phishing@pot@hotmail.com
Subject: Corvo
Date: Tue, 15 Aug 2023 15:25:40 +0000
Return-Path: natloulcinjd@hotmail.com

Full Email Body

Hello,

This is Leontyne Imai on behalf of the Student Loan Debt Department.

We tried to contact you at your home regarding your student loans.
Your student loans have been marked as eligible for forgiveness under the new 2023 guidelines.

Your case number is #55560, and your file will remain open for only one more day.

If you could please call your dedicated eligibility line at:
877-822-8290

Our office hours are 11am–7pm (EST), Monday–Friday.

Thank you,
Leontyne Imai

Important Note

The original email content was heavily encoded and obfuscated to bypass spam filters.
It has been cleaned and reconstructed for readability and analysis.

Phishing Indicators

- Suspicious/random sender email address
- Vague and meaningless subject ("Corvo")
- Obfuscated/encoded original content
- Financial lure (loan forgiveness offer)
- Urgency ("only one more day")
- Suspicious phone number
- Spam/advertising characteristics

Header Analysis

- SPF: Pass
- DKIM: Pass
- DMARC: Pass

Key Insight:
Even though authentication checks passed, the email is still malicious because attackers can use legitimate email services to send phishing messages.

Risk Classification - PHISHING / SPAM (High Risk)

Explanation

This email attempts to trick the user into calling a fraudulent number using a financial relief scam.
Attackers may then request sensitive information or payment, leading to identity theft or financial loss.

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5. Common Phishing Indicators

- Suspicious sender addresses
- Generic greetings
- Urgent or threatening language
- Suspicious links or phone numbers
- Encoded or poorly formatted content
- Unexpected financial offers

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6. Prevention Guidelines

What Users Should Do

- Verify sender email addresses carefully
- Hover over links before clicking

- Use official websites instead of email links
- Enable Two-Factor Authentication (2FA)
- Report suspicious emails immediately

What Users Should NOT Do

- Do not click unknown or suspicious links
- Do not download unexpected attachments
- Do not share passwords or OTPs
- Do not respond to suspicious emails

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7. Awareness Tips for Employees

- Always verify before trusting emails
- Be cautious of urgency and threats
- When unsure, report instead of clicking
- Understand that phishing can target anyone

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8. Conclusion

Phishing attacks remain one of the most common cybersecurity threats due to their reliance on human error.

This report demonstrates that even emails passing technical authentication checks (SPF, DKIM, DMARC) can still be malicious.

User awareness, combined with proper security practices, is the most effective defense against phishing attacks.

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9. References

- Public phishing datasets (GitHub)
- Email header analysis tools
- Cybersecurity awareness resources